

flea NEWS

37

Department of Entomology Iowa State University Ames, Iowa 50011-3222

Flea News Goes High Tech

With the advent of this issue, Flea News enters the age of high technology. In the future we intend to produce the copy by means of an Apple Macintosh SE computer and a desktop publishing program called Ready Set Go! As you can see, we have a new masthead, compliments of one of our more talented graduate students here in the Department of Entomology and we now have the capability of producing graphics as well as text, should the need arise. The original copy of this issue has been printed on an Apple LaserWriter IINT, and printed in the University printing plant. The type face is 12 point New Century Schoolbook for those of you with an interest in such matters.

Although we intend to adhere to the format followed in earlier issues, we hope to introduce some useful innovations as we become more accustomed to the technology. Suggestions from our readers are always welcomed, and if you have ideas as to how Flea News could be made more useful to you, please do not hesitate to put them forward.

Meanwhile, we wish to thank the many of you who have been so helpful in sending us photocopies and reprints. With your cooperation, we are accumulating what is certainly one of the best private libraries of literature dealing with Siphonaptera, including over 4500 reprints and over 100 bound volumes.

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Important Notice

Response has been quite good to our request for information to update our mailing list. At this writing, one hundred and fifteen responses have been received. However, our old list contained about 190 names and some of the people who have responded to our recent request were not on our old list. A certain amount of attrition is to be expected, but there are likely some who wish to continue to receive Flea News but have not yet responded. As with the last update, this issue is being sent to all names on the old mailing list whether we have heard from you or not. Copies of Flea News 38 will only be sent to persons who have already indicated their continuing interest, or those who respond to this request. Due to the dates of the national meetings of the Entomological Society of America, this issue is being sent out a bit earlier than is usual for the December number and the cutoff date for it is November 10th.

Happy Holidays!

DECEMBER 1988

News from our readers

One of the fringe benefits of the periodic update of our mailing list is that we frequently get news from people with whom we do not regularly correspond. In no particular order, following are excerpts from some of the returns. Mailing addresses of some of these respondents are given elsewhere.

F. G. A. M. Smit

"I am still 'full-time' researching life and works of the 17th and 18th century Dutch glass engravers. Recently published (self) a 200 page book on a particular engraver (after an incubation period of ten years)."

Norman G. Gratz

"I have retired over a year ago from the WHO and my post as Director of the Division of Vector Biology and Control. Since then I have been working a good deal of the time for the WHO and a number of other groups. Along with this I have been doing a good deal of writing and have a chapter in press with the CRC on "The Global Public Health Importance of Rodents" of which a good portion deals with flea-borne diseases, particularly plague and murine typhus.

"In addition I have started a large data-base and am well on the way to putting in the majority of my very large reprint collection dealing with most aspects of vector and rodent biology and control and the epidemiology of the vector and rodent-borne diseases."

Ralph P. Eckerlin

"I have just taken the editorship of the *Proceedings of the Helminthological Society of Washington* and I would like to encourage the submission of flea research papers to the *Proceedings*. I'd appreciate it if you would publish the enclosed announcement in a forthcoming Flea News." [See announcement elsewhere in this issue.]

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Abdu F. Azad

"Thanks for sending Flea News. By the way, do you know if anyone is rearing *Echidnophaga* or any *Ceratophyllus* sp.? If you do, please let me know." [In an earlier issue of Flea News I requested information about possible rearing colonies of fleas but got no response. Does anybody out there have colonies of anything other than *Ctenocephalides* or *Xenopsylla* ?]

Lance A. Durden

"...I am currently working on a number of papers, many of them on Anoplura, but some on fleas. ...Bob Traub and I are collaborating on a short chapter on louse and flea zoogeography as it relates to Sulawesi; this in connection with a conference I will be attending and presenting in London this September."

Frank J. Radovsky

"...for many years at the Bishop Museum in Honolulu, spent much of 1987 at Oregon State University where he taught medical entomology. During the time he was also heavily occupied with editing the *Journal of Medical Entomology*, following the acquisition of the journal as a publication of the Entomological Society of America. He is now Director of Research and Collections, as well as Assistant Director of the North Carolina State Museum of Natural Sciences, in Raleigh, starting in November, 1987. Much of his time is occupied with administration, exhibits development, and editing, but he is continuing research as time permits. The biology of *Tunga* fleas continues to be one active interest."

Cluff E. Hopla

Although he did not mention it on his return, Cluff is now one of the editors of the *Journal of Medical Entomology* of the Entomological Society of America.

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Presented Papers

Following is a listing of papers presented at scientific meetings noted since Flea News 36.

At the 3rd Annual Conference of the European Region of the Society for Vector Ecology, Pembroke College, Cambridge.

Use of indicators of sexual maturation and regression in fleas in host-specificity studies. B. Williams, Medical Entomology Centre at the University of Cambridge, Cambridge, UK.

At the XVIII International Congress of Entomology, Vancouver, British Columbia, July 3rd to 9th, 1988.

External morphology of flea larvae (Siphonaptera) and its significance in taxonomy. R. L. C. Pilgrim, New Zealand.

Survival and reproduction of artificially fed cat fleas, *Ctenocephalides felis* (Bouché) (Siphonaptera: Pulicidae). S. F. Wade* & J. R. Georgi, USA.

Water relationships of adult and immature cat fleas *Ctenocephalides felis* (Bouché) (Siphonaptera: Pulicidae). A. G. Appel, USA.

Development of cat fleas, *Ctenocephalides felis felis* on host animals. L. A. Lemke* & P. G. Koehler, USA.

The cat flea, *Ctenocephalides felis* as a potential vector of murine typhus. A. F. Azad, USA.

Flea allergies in pets. R. E. W. Halliwell, UK.

Current cat flea research in the United Kingdom. B. Williams, UK.

Study of *Ctenocephalides canis* and *Archaeopsylla erinacei erinacei* in Ireland. K. P. Baker* & R. Mulcahy, Ireland.

Studies on the bionomics and host relations of fleas of the genus *Ctenocephalides* in Tamil Nadu, India. S. A. Joseph, India.

Ectoparasite surveys in Iraq with emphasis on *Ctenocephalides* sp. J. Abul-Hab, Iraq.

The veterinary and medical importance of *Ctenocephalides* spe[cies] in Israel. Y. K. Mumcuoglu, Israel.

The genus *Ctenocephalides* in Israel - biology, importance and control. S. Rosen, A. Hadani, I. Yeruham* & L. M. Ziv, Israel.

Investigations on *Ctenocephalides* sp. in Nigeria. E. B. Otesile, Nigeria.

The status of the cat flea in Denmark and prospects for new methods of prevention and control. A. Olsen, Denmark.

The effects of synthetic pyrethroids on *Ctenocephalides felis* in Tanzania. B. S. Kilonzo* & D. N. Gisakanyi, Tanzania.

Bioassay technique for the evaluation of insecticides used in the control of *Ctenocephalides felis* (Bouché) (Siphonaptera: Pulicidae). W. A. Donahue, USA.

Insect growth regulators for the control of cat fleas. P. G. Koehler*, R. S. Patterson & L. A. Lemke, USA.

A film on the life cycle of the cat flea. K. Keldsen, Denmark.

Patterns of geographic distribution and

dispersal among North American cliff swallow fleas (Siphonaptera: Ceratophyllidae). T. D. Galloway, Canada.

The evolution of parasitism in arthropods. Yu. S. Balashov, USSR.

Parasitic arthropod-host immune interactions. S. K. Wikel, USA.

The structure of parasitic arthropod communities. K. C. Kim, USA.

Evolution of sensilla in parasitic arthropods. S. B. McIver, Canada.

Ectoparasites and cliff swallows: a study in coadaptation. C. E. Hopla, USA.

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Plague Update

The main thrust of Flea News has always been oriented more toward the bibliographical and nomenclatural than the veterinary and public health professions. Nevertheless, we do on occasion provide information on bubonic plague since it remains an important flea-transmitted disease of world-wide interest and distribution. In order to keep this information current and authoritative, we have solicited the aid of Dr. Allan M. Barnes, United States Public Health Service, Centers for Disease Control, Fort Collins, CO. Dr. Barnes is an acknowledged expert on plague and sends us the following information.

"Plague is a notifiable disease under the International Health Regulations of WHO and reported cases are enumerated in the Weekly Epidemiological Record (WER), WHO, Geneva, which serves as a primary source of information to other agencies. The Centers for Disease Control is the basic source for plague information in the United States and acts as a reference center and reporting agency to WHO. CDC information is summarized

in Morbidity and Mortality Weekly Report (MMWR) Surveillance Summaries, MMRW itself, and in occasional papers and reports.

"The number of plague cases in the United States this year to October 1 has reached 14. There have been no fatalities. The cases occurred in: Arizona (1), California (2), Colorado (4), New Mexico (6) and Texas (1). Animal plague activity during 1988 as detected by joint CDC-state surveillance has been low, although the number of human cases is about average for the past decade. A discussion of the 14 cases will appear in MMWR from CDC in the near future.

"A scan through the WHO Weekly Epidemiological Record for 1988 to date shows diminished plague activity around the world, following severe outbreaks in Africa and South America in immediately previous years. Nevertheless, plague cases were reported in 1988 from Bolivia, Brazil, and Peru, and from Madagascar and Viet Nam. As of September 29, 1988, Madagascar, Tanzania, Uganda and Zaire remained on the "Infected Areas" list in WER, as do Bolivia, Brazil, Ecuador and Peru. Madagascar has reported human plague in all but one year since 1968." AMB.

* * * * *

The Defense Pest Management Information Analysis Center warns that flea control products, particularly flea dips for pets, may contain cholinesterase-inhibiting organophosphate pesticides. At least two cases of human illness associated with flea dip products have been reported in the last two to three years. Workers should wear protective clothing, use proper dilutions of the product, and ensure proper clean-up of the work area and any product residues after treatment. It makes us wonder what the pet must be going through while being treated.

Mailing List

Additions

Dr. Arthur Appel, Department of Entomology, 301 Funchess Hall, Auburn University, Auburn, AL 36849-5413

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Rolan Tripp, DVM, 5231 Alhaina Drive, Woodland Hills, CA 91364

Kimito Uchikawa, Ph.D., Department of Parasitology, Shinshu University School of Medicine, Matsumoto 390, Japan

Dr. V. S. Vashchenok, Laboratory of Parasitology, Zoological Institute, Academy of Sciences USSR, Leningrad, 199034, U.S.S.R.

Dr. Wu Hou-yong, Department of Vector Biology & Control, Institute of Microbiology & Epidemiology, Academy of Military Medical Sciences, P. O. Box 130, Beijing, 100850, People's Republic of China

Deletions

Dr. W. L. Jellison, 504 South Third Street, Hamilton, MT 59840

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Reprints

Since Flea News 36, the following have supplied reprints and photocopies, for which we are most grateful: G. S. Pfaffenberger, V. S. Vashchenok, H. J. Ego-sque, A. Dudich, Z. S. Amr, P. M. Linardi, J. Segerman, E. Méndez, S. G. Medvedev, A. Kissileff, J. Müller, J. C. Beaucournu, W. Skuratowicz, Li Guizhen, Yu Xin & Ye Rui-yu. Such contributions become especially important when articles are published in small, obscure journals or in journals not available in the Iowa State University Library. We thank you all for your continued cooperation

The Proceedings of the Helminthological Society of Washington solicits flea research manuscripts for publication. Research dealing with flea taxonomy, systematics, morphology, ecology, biogeography (including surveys), life histories, etc. are welcome and would fit into the scope of the papers usually published.

Don't let the title fool you! The Proceedings is not a proceedings of the monthly meetings of the Helminthological Society of Washington. It is a peer reviewed journal of high quality and international distribution. Currently papers on helminths, but also protozoans, leeches, parasitic crustaceans and mites are being published. The last flea taxon described in the Proceedings was *Ctenophthalmus tholatus* Traub, 1963.

The Proceedings have been published continuously since 1934 with two issues annually in January and July. There is no backlog of manuscripts and well written papers can be published quickly (4-6 months). Page charges to members of the Society are \$40 per page (\$80 for non-members). Membership costs only \$15 annually. See a recent issue for format.

Send manuscripts for consideration to:

Ralph P. Eckerlin, Editor
Proceedings of the Helminthological
Society of Washington
Natural Sciences Division
Northern Virginia Community College
Annandale, VA 22003

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Fleas in Literature

Following is a continuation of quotations from some early authors prepared by F. G. A. M. Smit. Readers are referred to Flea News 35, p. 259-261, for the first installment which includes the original references.

Copulation

They copulate, the male ascending upon the female as Flies doe, and they both goe, leap, and rest together. They stick long together, and are hardly pulled asunder. (Moufet 1634: 276; 1658: 1102)

Gravid females

After copulation presently almost, the female full of Egges seems fatter; which though in her belly they seem long, very small, very many, and white, yet when they are layd, they turn presently black, and turn into littles Fleas that bite most cruelly (if we may grant what Pennius saith). (Moufet 1634: 276; 1658: 1102)

Metamorphosis

Aristotle thinks, that from them, be they Egges, Nits, or little Worms, no other Creature breeds, and I should willingly subscribe to him, but that I think Nature made nothing in vain. (Moufet 1634: 276; 1658: 1102)

Jumping

The hind legs are distinctly larger than the other legs for the purpose of jumping which is the result of the stretching of the bent legs. (Aristoteles ±300 BC)

Fleas jump with their hind feet. (Plinius 77: Liber XI Chapter 33)

They leap rather than fly. (Cantimpré 1233-1248)

A flea is very light and escapes dangers by jumping, not walking. (B. Anglicus 1483)

Prior to jumping a flea crouches with the head on the ground, as well as the two long hind legs; the flea then shakes rapidly and at once raises its head and jumps a long distance; it does not fly as does the cricket commonly. (Severinus 1645)

Origin

When a flea is divided into two parts it comes to life again. (Cantimpré 1233-1248; Anonymous 1491).

Provisional list of taxa described in 1987

Caenopsylla janineae Beaucournu & Gouat
Caenopsylla laptevi ibera Beaucournu & Marquez
Amphipsylla longispina gongheensis Zhang & Ma
Stenischia xiei Li
Plocopsylla viracocha Schramm & Lewis
Plocopsylla nungui Schramm & Lewis
Plocopsylla athena Schramm & Lewis
Plocopsylla kilya Schramm & Lewis
Frontopsylla (F.) *elatoides intermedia* Cai, Wu & Zhang
Geusibia falsotorosa Wu, Ni & Li
Ceratophyllus (*Monopsyllus*) *hamatus* Cai & Wu
Ceratophyllus (*Monopsyllus*) *forficis* Cai & Wu
Macrostylophora nandanensis Li, Zeng & Zeng
Amphalius clarus tianshanensis Yu, Ye & Xie
Ectinorus (*Ectinorus*) *nomisis* Smit
Ectinorus (*Ichyonus*) *onychius deplexus* Smit
Parapsyllus ambrosius Smit
Tiamastus deflatus Smit
Polygenis (*Gephyropsylla*) *klagesi rangeli* Smit
Polygenis (*Polygenis*) *trapidoi mendezi* Smit
Paradoxopsyllus rhombomysus Li, Huang & Sun
Paradoxopsyllus inferioprocerus Yu, et al.
Wagnerina liui Yu

Provisional list of taxa described in 1988

Amphipsylla apiciflata Liu, Xu & Li
Paradoxopsyllus aculeolatus Ge & Ma
Amphipsylla yadongensis Wang & Wang
Plocopsylla lewisi Beaucournu & Gallardo
Ctenoparia propinqua Beaucournu & Gallardo
Callopsylla arcuata Ge, Wang & Ma
Nycteridopsylla liui Wu, Chen & Liu
Metastivalius novaehiberniae Beaucournu & Mahnert

Requested by E-mail?

— Have
○ need
Dr. William L. Jellison, having retired from the U. S. Public Health Service, has deposited remaining copies of the following publications at Iowa State University. Persons interested in obtaining copies should write Flea News.

— *Opisodasys* Jordan, 1933, a genus of Siphonaptera. J. Parasitol. 25(5): 413-420 (1939).

— Review - Fleas of eastern United States. J. Parasitol. 26(4): 335-366 (1940).
Siphonaptera. Species and host list of

Montana fleas. Misl. Publ. 2 Mont. Bd. Ent. 22 p. (1943). with Kohls & Mills.

— Siphonaptera: the genus *Oropsylla* in North America. J. Parasitol. 31(2): 83-97 (1945).

— Siphonaptera: a new species of *Conorhinopsylla* from Kansas. J. Kans. Ent. Soc. 18(3): 109-111 (1945).

— Siphonaptera: host distribution of the genus *Opisocrostitis* Jordan. Trans. Amer. Micr. Soc. 66(1): 64-69 (1947).

Flea Index Supplement 2. 1951-1960. 406 p. with Glesne (1967).

Fleas of Montana. Mt. Agric. Exp. Station Res. Rept. 29. 75+ p., with Senger (1973).

Fleas of western North America except Montana in the Rocky Mountain Laboratory collection. In: H. C. Taylor & J. Clark - Papers in honor of Jerry Flora. Bellingham, Wash., 55-136 (1976). with Senger. (This material has since been sent to the United States National Museum.)



Literature on Siphonaptera

1974 (List 20)

Kulkarni, S. M., H. R. Bhat & V. Dhanda
A survey of haematophagous arthropods in western Himalayas, Sikkim and hill districts of West Bengal. Fleas (Siphonaptera). Ind. J. med. Res 62(7): 1061-1088.

1975 (List 22)

Ito, S., J. W. Vinson & T. J. McGuire, Jr. - Murine typhus rickettsiae in oriental rat flea. Ann. N. Y. Acad. Sci. 266: 35-60.

1976 (list 20)

Marks, G. & W. K. Beatty - Epidemics. Charles Scribner's Sons, New York. 323 pp.

1979 (List 19)

Jones, A. M., J. Mann & R. Brazier - Human plague in New Mexico: report of three autopsied cases. J. Forensic Sci. 24: 26-38.

1980 (List 16)

Li G. [K.-c.] - Present studies on *Para-*

doxopsyllus - classification, distribution, morphology and parasites. Bull. Guizhou Med. Coll. 5(1): 1-7 [In Chinese, no English summary]

1981 (List 14)

*Adrian, W. J. - Manual of common wildlife diseases in Colorado. Colorado Division of Wildlife. 139 pp.

1982 (List 11)

*Tigertt, W. D. - Plague. In: Evans, A. S. & H. A. Feldman (Eds.), Bacterial Infections of Humans. Plenum Medical Book Co., New York, pp. 403-415.

1983 (List 9)

Haas, G. E., A. J. Beck & P. Q. Tomich - Bat fleas (Siphonaptera: Ischnopsyllidae) in California. Bull. Sthn. Calif. Acad. Sci. 82(3): 103-114.

*Mann, J. - Plague. Current Therapy in Infectious Disease. 1983-1984.

Müller, J. - Mitarbeit erbeten bei der Erfassung von Ektoparasiten der Wildsäuger. Säugetierkd. Inform. 2(7): 81-82.

1984 (List 9)

Li G. [K.-c.] - Fleas (*Paradoxopsyllus*) in Guizhou Province: their relationship with hosts and characteristics of animal geology. Development of Science in Guizhou Province, pp. 205-212. [In Chinese, no English summary]

*Mann, J. - Plague - perspective on a rare disease. Western J. Med. 140: 650-651.

1985 (List 7)

*Diallo, J. S., Ed. - Manifestations oph-

talmologiques des parasitoses. Masson, Paris, France. 358 pp. [Siph. reportedly in Chapter 1 of Section 4]

Dudich, A. - Ectoparasites (Siphonaptera) of the small mammals in the Burda Mountains (Kováčovské kopce). Spravodja Oblastného Podunajského Múzea v Komárne Prírodné Vedy 5: 127-140.

Mollaret, H. H. & J. Brossollet - Alexandre Yersin, la vainqueur de la peste. Fayard, Paris. 320 pp.

Müller, J. & E. Leuthold - Flöhe (Siphonaptera, Ischnopsyllidae) und Fliegen (Diptera, Nycteribiidae) von Fledermäusen aus dem Bezirk Magdeburg und Nachbargebieten. Abh. Ber. Naturkd. Vorgesch. 12(6): 36-46.

1986 (List 5)

Aebli-Wortinger, A. - Fleas and related health problems in dogs and cats. Vet. Technician 7(6): 278-280.

Beaucournu, J.-C. & J. C. Torres-Mura - Un *Tetrapsyllus* nouveau du Chili (Siphonaptera, Rhopalopsyllidae). Revue fr. Ent., (N.S.) 8(1): 9-12. [*Tetrapsyllus satyrus*]

*Brown, T. L. - Plague prevention: How to avoid becoming New Mexico's next human plague case. New Mexico Epidemiology Report.

Carmichael, A. G. - Plague and the poor in Renaissance Florence. Cambridge University Press, Cambridge & New York. 180 pp.

Chauvancy, G. & J. M. Duplantier - Un nouveau modèle de piège pliant pour la capture de petits rongeurs. Mammalia 50(2): 272-274.

Downes, A. & R. Glen - George Pearson

Holland 1911-1985. Bull. ent. Soc. Canada 18(2): 71-75.

Kharitonova, N. N. & Yu. A. Leonov - Omsk hemorrhagic fever. Ecology of the agent and epizootology. Oxonian Press PVT Ltd., New Delhi. Trans. of Omskaya Gemorragicheskaya Likhoradka, publ. in Russian, Naukna Publishers, Novosibirsk, USSR Acad. Sci. Siberian Dep., Biol. Inst.

Müller, J. - Das aktuelle Flohartenspektrum (Siphonaptera) auf Stadthunden. Wiss. Z. Karl-Marx-Univ. Leipzig. Math.-Naturwiss. R. 35(6): 653-659.

O'Farrel, T. P. & L. Gilbertson - Ecology of the Desert kit fox, *Vulpes macrotus arsipus*, in the Mojave Desert of southern California. Bull. Sthn. Calif. Acad. Sci. 85(1): 1-15.

Reedy, L. M. - Common parasitic problems in small animal dermatology. J. Amer. vet. Med. Assoc. 188(4): 362-364.

Sheremet, O. V., V. N. Milyutin, Ya. N. Korganov & V. A. Kopylov - Virulence, viability and antibiotic sensitivity of *Yersinia pestis* grown in some culture media at 28° C and 37° C. Zh. Mikrobiol., Moskva (5): 53-56.

Williams, J. E., L. Arntzen, G. L. Tyn-dal & M. Isaäcson - Application of enzyme immunoassays for the confirmation of clinically suspect plague in Namibia, 1982. Bull. Wrld Hlth Org. 64 (5): 745-752.

*Williams, R. E. - Epidemiology and control of ectoparasites of swine. Vet. Clinics of N. Amer., Food Anim. Practice 2(2): 469-480.

Ye R-y & Yu X - Studies on flea specimens of dorso-ventral posture. Acta ent. Sinica 29(4): 436-445.

Ye R-y & Yu X - Studies on the shape of egg burster of first instar larvae of flea and its taxonomic significance. Reports on Local Diseases 1(2): 107-111.

1987 (List 3)

Abdullah, I. A. & I. S. Hassan - Ectoparasites of the long-eared hedgehog *Hemichinus auritus* Gmelin, in Ninevah district, Iraq. J. biol. Sci. Res. 18(2): 43-52.

Aseeva, L. E., L. A. Shevchenko, N. Ya. Shimanyuk, B. D. Rublev & B. N. Mishankin - Chemiluminescence in the evaluation of the modulating action of *Yersinia pestis* adenylate cyclase on peritoneal leukocytes of guinea pigs. Zh. Mikrobiol., Moskva (7): 59-63.

Beaucournu, J. C. & H. Launay - Siphonapteres et taxonomie. Bull. Soc. Fr. Parasit. 5: 133-144.

Beaucournu, J. C. & F. J. Marquez - *Caenopsylla laptevi ibera* ssp. nova (Siphonaptera, Leptopsyllidae) est la *Caenopsylla* inféodée au lapin de garenne dans la Péninsule Ibérique. Rev. Iber. Parasit. 47(1): 93-99.

Becker, E. C. - Publications of G. P. Holland. Can. Field-nat. 101: 472-473.

Berseth, W. C. & P. Zubac - Habitat associations of fleas, Siphonaptera, parasitizing the Short-tailed shrew, *Blarina brevicauda*. Can. Field-nat. 101(4): 594-596.

Bicak, L. J. - Pestilence. Amer. Biol. Teacher 49: 99-102.

Bork, K., K. Honomichl & N. Hoede - Flohbisse durch *Archaeopsylla erinacei*, den Igelfloh. Hautarzt 38(11): 690-692.

Bryukhanova, L. V. - Comparison of ecological peculiarities of the fleas *Citel-*

lophilus tesquorum (Wagn., 1898) and *Neopsylla setosa* (Wagn., 1898), abundant parasites of the little souslik. Byul. Moskov. Obshch. ispyt Prirod., Biol. 92 (5): 70-74 [Brief English summary].

Burdelov, L. A. - Crisis in plague epizootology: causes and the way of overcoming. Zh. Obshch. Biol. 48(6): 816-827.

Carmen, G. M. - A tribute to George Pearson Holland. Can. Field-nat. 101: 470-472.

Cherkasova, T. D., P. R. Vengrov, V. I. Melikhov, V. P. Avrorov & V. A. Yurkiv - Cyclic nucleotide content in the course of the development of toxoinfectious shock caused by *Yersinia pestis* "mouse" toxin. Zh. Mikrobiol., Moskva (7): 70-74.

Cyprich, D. & M. Krumpal - Doc. Ing. Milan Jurik, CSc., in memoriam. Biologia, Bratislava 42(10): 1024 [In Czech].

Delaporte, F. - Book review - Alexandre Yersin, la vainqueur de la peste by H. H. Mollaret & J. Brossollet. Fayard, Paris. 1985. 320 pp. Bull. Hist. Med. 61(2): 288-289.

Dudich, A. - A contribution to the knowledge of fleas (Insecta: Siphonaptera) on small mammals in the Trábeč mountains. Ochrana Přírody 8: 277-289.

Dudich, A., M. Ambros, A. Štollmann & J. Kováčik - Ectoparasitic arthropods on small mammals in the slovak karst protected landscape area. 1. Plešivská Planina and Koniar Plateaus. Ochrana Přírody 8: 101-121.

Galun, R. - Regulation of blood gorging. Insect Sci. Applic. 8(4/5/6): 623-624.

Haitlinger, R. - Arthropods (Siphonaptera, Anoplura, Acari) occurring in Poland on *Spermophilus suslicus* (Guel-

denstaedt, 1770)(Mammalia: Rodentia). Wiad. Parasit. 33(6): 701-705.

*Hatfield, P. R. - An immunological approach to vector control: how host [mouse] immune responses to vector antigens may affect the mosquito *Aedes aegypti* (Linnaeus) as a vector of yellow fever virus, and the flea *Xenopsylla cheopis*. Thesis, London School of Hygiene and Tropical Medicine. UK Index to Theses 36(1): 288.

Horak, I. G. - Arthropod parasites of some wild animals in South Africa and Namibia. J. S. Afr. Vet. Assoc. 58(4): 207-211.

Isin, Zh. M. & T. I. Tugambaev - Structural and functional properties and biological activity of *Yersinia pestis* capsular antigen, "murine" toxin, and endotoxin. Zh. Mikrobiol., Moskva (4): 91-98. [No English summary]

Kral, B. - Parasitenbefall in einem Tierheim. Wiener Tierärztl. Monatssch. 74(2): 71. [Abstr. of Dissertation]

Kratsov, A. N., V. Khodova, V. P. Zyuzina, V. I. Tynyanova & B. D. Rublev - Role of iron in the multiplication of *Yersinia pestis* in normal mammalian blood sera. Zh. Mikrobiol., Moskva (3): 12-16.

Le Boulenger, E. & P. Y. le Boulenger-Nguyen - A cost-efficient live trap for small mammals. Acta Theriol. 32(10): 140-144.

Ledger, J. A. - Medical importance of gamebird ectoparasites in southern Africa. S. Afr. J. Wildl. Res. 1987, Suppl. 1: 32-35.

Levi, M. I., T. A. Shimko & G. A. Temiralieva - Field trial of the enzyme immunoassay with the use of betalactamase as

a marker of monoclonal antibodies to *Yersinia pestis* capsular antigen. Zh. Mikrobiol., Moskva (3): 43-49.

Li G [K-c], Huang G-p & Sun Q - A new species of the genus *Paradoxopsyllus* Miyajima & Koidzumi, 1909 from Bayannur, Inner Mongolia. Reports on Local Diseases 2(4): 19-21. [*Paradoxopsyllus rhombomysus*]

Linardi, P. M. - Subsídio taxonômicos e filogenéticos obtidos por comparações faunísticas da distribuição geográfica dos sifonápteros ropalopsilinos. Rev. Brasil. Biol. 47(3): 397-407.

Lofflin, J. - Fleas and ticks: the never-ending story. Vet. Economics 28(4): 88.....104.

Manolov, A. D., V. P. Vasilyev, V. I. Zakrevsky, E. A. Snegirev & G. M. Kalei - Preparation of ¹²⁵I-labeled *Yersinia pestis* capsular antigen. Zh. Mikrobiol., Moskva (7): 117-118 [No English summary]

Marquez, F. J. & R. C. Soriguer - Intraproblational morphological variation in females of *Ctenophthalmus apertus meylani* Beaucournu, Gilot & Vericad, 1973. (Siphonaptera: Hystrichopsyllidae). Rev.ta iber. Parasit. 47(4): 419-424.

Melville, R. V. & J. D. D. Smith - Official lists and indexes of names and works in Zoology. Published by the International Trust for Zoological Nomenclature, c/o British Museum (Natural History), Cromwell Road, London SW7 5BD, U.K. on behalf of the International Commission on Zoological Nomenclature, 366 pp.

Méndez, E. - Elementos de la Fauna Panameña. Impreso en Panama. v-ix, 1-216 [cat flea: 147-149]

Palmer, R. - Book Review - Plague and the poor in Renaissance Florence. by Ann G. Carmichael. Cambridge University Press, Cambridge & New York. 1986. 180 pp. Bull. Hist. Med. 61(1): 122-123.

Pant, C. - Vector-borne diseases of man and their socio-economic impact. Insect Sci. Applic. 8 (4/5/6): 655-664.

Pfaffenberger, G. S. & V. B. Valencia - Parasites of public health concern associated with New Mexico desert cottontails (*Sylvilagus audubonii*) and blacktail jack rabbits (*Lepus californicus*). New Mexico J. Sci. 27(2): 79-85.

Picken, L. G., J. F. Carrol & A. F. Azad - Electrophysiological studies of the spectral sensitivities of cat fleas, *Ctenocephalides felis*, and oriental rat fleas, *Xenopsylla cheopis*, to monochromatic light. Entomologia exper. et appl. 45(2): 193-195.

*Prasad, R. S. - Nutrition and reproduction in haematophagous arthropods. Proc. Indian Acad. Sci., Anim. Sci. 96(3): 253-273.

Sadykov, A. D., D. S. Suierkulov & V. F. Bakhvalov - State and prospects of the serological study of osseous remains and excrements of carnivorous birds for the presence of *Yersinia pestis* capsular antigen on the territory of the Kirghiz SSR. Zh. Mikrobiol., Moskva (8): 109-110.

Shen Y-c, Yu X & Cao H-l - First discovery of *Frontopsylla* (*Frontopsylla*) *tianshanica* Schwarz, 1953 in China. Reports on Local Diseases 2(1): 44-45.

Sheremet, O. V., V. N. Milyutin, A. N. Terentyev & L. N. Morozova - Immunogenicity of *Yersinia pestis* grown in some culture media at 28° C and 37° C. Zh. Mikrobiol., Moskva (7): 63-68.

Sheremet, O. V., V. N. Neklyayev, V. A. Kopylov & A. A. Kanchukh - Antigenic composition of *Yersinia pestis* on some nutrient media at 28° C and 37° C. Zh. Mikrobiol., Moskva (4): 111-112.

Sheremet, O. V., A. N. Terentyev, I. N. Shatova & L. N. Morozova - Content of fraction 1 and VW-antigen in *Yersinia pestis* cultures grown in yeast-casein medium, yeast medium with Hottinger digest, and yeast medium with sunflower-seed protein. Zh. Mikrobiol., Moskva (6): 18-22.

Shevchenko, L. A. & B. N. Mishankin - *Yersinia pestis* adenylate cyclase: purification and some of its properties. Zh. Mikrobiol., Moskva (12): 15-19.

Shilov, M. N. & S. N. Varshavsky - Population of *Rhombomys opimus* (Rodentia, Cricetidae) in Cis-Ust-Urt and North Ust-Urt region, its association with plague epizooties. Zool. Zh. 66(10): 1552-1560.

Štollmann, A. & A. Dudich - Study material concerning small mammals (Insectivora, Rodentia) and their ectoparasites (Acarina, Anoplura, Siphonaptera) from the territory of the Slovak Socialist Republic. 2. Survey of sites from the West-Slovak region. Ochrana Prírody 8: 55-69.

Tonelli, E. A. - Utilizacion experimental del D-Limoneno en forma de bañaciones para el tratamiento de infestaciones por pulgas y garrapatas en pequeños animales. Vet. Argentina 4 (40): 931-937.

Toporkov, V. P., M. I. Levi, R. A. Beloborodov, Sh. N. Zaropov & M. A. Afanasyeva - Possibility of the detection of *Yersinia pestis* in nature after its transition into the reservation phase. Zh. Mikrobiol., Moskva (4): 113-114.

*Valencia, V. B. - Survey of the metazoan parasites from sympatric *Sylvilagus audubonii* Nelson and *Lepus californicus* Mearns in eastern New Mexico. M.S. Thesis, Eastern New Mexico University, Portales, NM, 94 pp.

Ye R-y & Yu X - Taxonomic discussion on *Citellophilus tesquorum deztyuensis* Mikulin, 1951 (Siphonaptera: Ceratophyllidae). Reports on Local Diseases 2(3): 25-30.

Ye R-y, Yu X & Huang G-m - The cases of flea teratology in Xinjiang (2). Acta ent. Sinica 30(2): 225-228.

Yu X - Description of a new species of *Wanerina*. (Siphonaptera: Hystrichopsyllidae). Reports on Local Diseases 2(4): 26-28. [W. liui]

Yu X, Ye R-y & Xie X-c - A new subspecies of *Amphalius* (Siphonaptera: Ceratophyllidae). Acta Zootax. Sinica. 12(4): 418-420. [*Amphalius clarus tianshanensis*]

Zuger, A. & S. H. Miles - Physicians, AIDS, and occupational risk. Historic traditions and ethical obligations. JAMA 258(14): 1924-1928. [Plague: 1924-1925]

1988 (List 2)

Anonymous - Beware: Flea & tick products implicated in cancer and the poisonings of companion animals. PETA [People for the Ethical Treatment of Animals], 2 pp.

Anonymous - Concern about safe use of flea control products. J. Amer. Vet. Med. Assoc. 193(7): 758.

Anonymous - Organophosphate toxicity associated with flea-dip products - Cali-

fornia. Morb. Mort. Wkly. Rep. 37(21): 329-330.

Anonymous - Human plague - United States, 1988. Morb. Mort. Wkly. Rept. 37(42): 653-656.

Beard, M. L., G. O. Maupin, A. M. Barnes & E. F. Marshall - Laboratory trials of Cholecalciferol against *Spermophilus variegatus* (Rock Squirrels), a source of human plague (*Yersinia pestis*) in the southwestern United States. J. environ. Hlth. 50(5): 287-289.

Beaucournu, J. C. & M. Gallardo - Puc- es nouvelles d'Argentine (Insecta: Siphonaptera). Rev. suisse Zool. 95(1): 99-112. [*Plocopsylla lewisi*, *Ctenoparia propinqua*]

Beaucournu, J. C., H. Launay & A. Sklair - Les anomalies des spermathèques et des conduita génitiaux chez les Siphonaptères (Insecta). Ann. Parasitol. Hum. Comp. 63: 64-75.

Beaucournu, J. C. & V. Mahnert - Un *Metastivalius* nouveau de Nouvelle-Irlande (Siphonaptera: Pygiopsyllidae). Rev. suisse Zool. 95(2): 533-538. [*M. novae-ahiberniae*]

Bell, P. J., H. R. Burton & J. A. van Franeker - Aspects of the biology of *Glaciopsyllus antarcticus* (Siphonaptera: Ceratophyllidae) during the breeding season of the host (*Fulmarus glacialis*). Polar Biology 8(6): 403-410.

Best, T. L. - *Dipodomys spectabilis*. Mammal. Species 311: 1-10. [Siph.: 5]

Boag, B. - Observations on the seasonal incidence of myxomatosis and its interactions with helminth parasites in the European rabbit (*Oryctolagus cuniculus*). J. Wildl. Dis. 24(3): 450-455.

Bronner, G., S. Gordon & J. Meester - *Otomys irroratus*. Mammal. Species 308: 1-6. [Siph: 5]

Brown, A. E., S. R. Meek, N. Maneechai & G. E. Lewis - Murine typhus among Khmers living at an evacuation site on the Thai-Kampuchean border. Amer. J. Trop. Med. Hyg. 38(1): 168-171.

Bryon, D. & W. Robinson - Locating flea hot spots. Pest Control 56(6): 28-30.

Chamberlain, W. F., J. Maciejewska & J. J. Matter - Response of the larvae and pupae of the Oriental rat flea (Siphonaptera: Pulicidae) to chemicals of different chemical types. J. econ. Ent. 81(5): 1420-1425.

Ding X-l, Li B-s & Liu J-x - A preliminary study on distribution of three subspecies of *Ophthalmopsylla praefecta* in Gansu Province, China. Acta zootax. Sinica 13(2): 178-181.

Durden, L. A. - Review - A colour atlas of insect tissues via the flea. Bull. amat. Ent. Soc. 47: 138-140.

Durden, L. A., D. A. Lott & B. Ingram - Fleas (Siphonaptera) from a dormouse nest in Devon. Ent. mon. Mag. 124: 56.

Eckerlin, R. P. - Review - The Rothschild collection of fleas. The Ceratophyllidae... J. Parasit. 74(2): 352.

El-Gazzar, L., R. S. Patterson & P. G. Koehler - Comparisons of cat flea (Siphonaptera: Pulicidae) adult and larval insecticide susceptibility. Fla. Entomol. 71(3): 359-363.

Gannon, W. L. - *Zapus trinotatus*. Mammal. Species 315: 1-5. [Siph: 3]

Ge L & Ma L-m - A new species of the genus *Paradoxopsyllus* from Xizang, Chi-

na (Siphonaptera: Leptopsyllidae). Acta zootax. Sinica 13(1): 91-92. [*P. aculeolatus*]

Ge L., Wang C-g & Ma L-m - A new species of the genus *Callopsylla* from Xizang (Siphonaptera: Ceratophyllidae). Acta zootax. Sinica 13(2): 175-177. [*C. arcuata*]

Gold, E. Yu - From the history of plague control. Zh. Mikrobiol., Moskva (4): 120-124. [No English summary]

Hensley, A. P. & K. T. Wilkins - *Leptoncyteris nivalis*. Mammal. Species 307: 1-4. [Siph: 3]

Khrutskii, A. E., V. Ya. Linnik, I. T. Arzamasov, A. G. Lebedskaya & V. F. Litvinov - Controlling ectoparasites of mink. Veterinariya, Moskva (5): 38-39.

Kocianová, E. & O. Kožuch - A contribution to the parasite fauna in winter nests of the common mole (*Talpa europaea* L.) and incidents of its infection with tick-borne encephalitis virus (TBE) and rickettsia *Coxiella burnetii*. Folia Parasit. 35: 175-180.

Kozlov, M. P., V. P. Nadeina & I. V. Chumakova - Haemolymph cells of fleas and their phagocytic activity. Parazitologiya 22(4): 321-328.

Krampitz, H. E. & A. Haberkorn - Experimental treatment of *Hepatozoon* infections with the anticoccidial agent Toltrazuril. J. Vet. Med. B 35: 131-137.

Kulišić, Z. - Ectoparasites infesting pigeons (*Columba livia*) detected in the city of Belgrad. Acta Veterin. 38(2-3): 139-144.

Lewis, R. E., J. H. Lewis & C. Maser - The fleas of the Pacific Northwest. Oregon State University Press, Corvallis, OR 296 pp.

Li W-d & Yu X - A new record of *Ischnopsyllus* from China. Acta zootax. Sinica 13(1): 80. [*I. octactenus* (Kolenati, 1856)]

Liu Q, Xu G-s & Li Z-y - Description of a new species of *Amphipsylla* from Nei Mongol, China (Siphonaptera: Leptopsyllidae). Acta zootax. Sinica 13(1): 88-90. [*A. apiciflata*]

Ma L-m - Abundance of fleas in relation to population fluctuations of their hosts. Acta ent. sin. 31(1): 50-54.

Macêdo, R. H. & M. A. Mares - *Neotoma albigula*. Mammal. Species 310: 1-7. [Siph: 4]

Méndez, E. - Parasites of vampire bats. Chapter 13, pp. 191-206. In: Natural history of vampire bats. Greenhall, A. M. & U. Schmidt (Eds.) CRC Press, Inc., Boca Raton, FL 246 pp.

Mix, J. - Think change when thinking fleas. Pest Control 56(6): 35-36.

Oliver, J. H., Jr. - Crisis in biosystematics of arthropods. Science 240(4855): 967.

Ostrowski, S. R. - Concerned about safe use of flea control products. J. Amer. Vet. Med. Assoc. 193(7): 758-759.

Phillips, A. P., B. C. Morris, D. Hall, M. Glenister & J. E. Williams - Identification of encapsulated and non-encapsulated *Yersinia pestis* by immunofluorescence tests using polyclonal and monoclonal antibodies. Epidem. Inf. 101(1): 59-73.

Punsky, E. E., M. I. Levi, L. A. Traub, I. E. Punsky & G. A. Temiralieva - Antigennuria in plague-infected great gerbils. Zh. Mikrobiol., Moskva (2): 112-118.

Scheidt, V. J. - Flea allergy dermatitis. Small Animal Practice 18(5): 1023-1042.

Segerman, J. & L. E. O. Braack - New records of bat fleas (Siphonaptera) from South Africa. J. ent. Soc. sth. Afr. 51(1): 149-150.

Skuratowicz, W. - *Megabothris walkeri* (Rothschild, 1902) (Siphonaptera: Ceratophyllidae) from Poland. Frag. Faun. 31(14): 411-428.

Skuratowicz, W. - Fleas (Siphonaptera: Ischnopsyllidae, Ceratophyllidae) from bats (Chiroptera) collected in Poland. Frag. Faun. 31(15): 429-444.

Smith, S. A. & M. E. Clay - Biological and morphological studies on the bat flea, *Myodopsylla insignis* (Siphonaptera: Ischnopsyllidae). J. med. Ent. 25(5): 413-424.

Ubico, S. R., G. O. Maupin, K. A. Fagerstone & R. G. McLean - A plague epizootic in the white-tailed prairie dogs (*Cynomys leucurus*) of Meeteetse, Wyoming. J. Wildl. Dis. 24(3): 399-406.

Vashchenok, V. S. - Fleas - vectors of pathogens causing illness in humans and animals. Trudy Akad. Nauk SSSR Zool. Inst. Vol. 166, "Nauka", Leningrad, 160 pp.

Vashchenok, V. S., R. S. Karandina & L. V. Bryukhanova - Blood amount engorged by fleas of different species under experiment. Parazitologiya 22(4): 312-320.

Wang C-g & Wang S-r - A new species of *Amphipsylla* from Xizang, China (Siphonaptera: Leptopsyllidae). Acta zootax. Sinica 13(1): 93-94. [*A. yadongensis*]

Wang C-g & Wang S-r - A new record of *Palaeopsylla* from China. Acta zootax.

Sinica. 13(1): 56.

Wang, D-q - A tentative classification of the flea larvae (Siphonaptera). Acta ent. Sinica 31(3): 326-331.

Whittaker, J. O., Jr. & T. W. French - Ectoparasites and other arthropod associates of the hairy-tailed mole, *Parascalops breweri*. Great Lakes Ent. 21(1): 39-41.

Whittaker, J. O., Jr. & T. W. French - Ectoparasites and other arthropod associates of some voles and shrews from the Catskill Mountains of New York. Great Lakes Ent. 21(1): 43-48.

Williams, B. & T. E. Hallas - *Cheyletus eruditus* (Schrank) (Acari) feeding on eggs of the cat flea *Ctenocephalides felis* (Bouché) (Siphonaptera). Ent. mon. Mag. 124: 144.

Williams, J. E., M. K. Gentry, C. A. Braden, G. L. Tyndall, P. L. Altieri, S. Berman & D. M. Robinson - A monoclonal antibody for the specific diagnosis of plague. Bull. W.H.O. 66(1): 77-82.

Wu H-y, Chen J-x & Liu Q - Description of a new species of *Nycteridopsylla* (Siphonaptera: Ischnopsyllidae). Acta zootax. Sinica 13(2): 172-174. [*N. liui*]

Wu W-z - A new record of [a] flea from China. Acta ent. Sinica 31(2): 227.

Xiao B-l & Wang D-q - Two flea larvae of *Marmota himalayana*. Acta ent. Sinica 31(1): 96-98.

Yanni, J. M., R. E. W. Halliwell & C. H. Tracy - Effect of AHR-5333 on flea antigen extract-induced skin reactions in flea-allergic dogs. J. Vet. Pharmacology & Therapeutics 11(2): 221-225.



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Flea News is mainly a taxonomic and bibliographic newsletter that is produced semiannually by Robert E. and Joanne H. Lewis of the Department of Entomology, Iowa State University, Ames, IA 50011-3222 USA. It is sent free of charge to those persons interested in the Siphonaptera, both on a theoretical and an applied basis. Recipients of **Flea News** are asked to send reprints of their works, if available, as well as news items that they deem appropriate. The co-editors have attempted to be as accurate in the editing process as possible, but we blame the machinery for any errors that have escaped our attention. Anyone who has ever attempted "Desk-top Publishing" for the first time will know and understand.

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